PROJECT PROPOSAL

Cloud Based Big Data Processing and Predictive Analytics for 5G O RAN Networks

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Course	Big Data Analytics & Cloud Computing
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1. Project Motivation

This proposal presents a general plan for using cloud computing and big data tools to study data from a 5G Open Radio Access Network. The project uses the NetsLab 5G O RAN IDD dataset, which was collected from a physical 5G Open RAN testbed at University College Dublin. The dataset includes network traffic records and radio measurements from user equipment. Examples of the radio measurements are RSRP, RSRQ, RSSI, SINR, CQI, uplink and downlink MCS, BLER, and cell information [1] [2].

The main purpose is to explore how a cloud based data pipeline can store, clean, combine, and analyze different types of 5G network data. A prediction model is also included. The main candidate prediction is whether a network record represents a normal state or an abnormal state. A more detailed prediction, such as the type of abnormal activity, may be explored in the project.

2. Background and Importance

Modern 5G networks create data at several layers. The radio layer describes the connection between user equipment and the radio network. The network layer describes traffic, connections, packets, and services. Looking at only one layer may hide useful information. For example, a large increase in traffic may be caused by normal demand, poor radio conditions, or unusual activity. Combining radio and traffic information may give a clearer view of the network state.

This issue is important because manual analysis becomes difficult when many users, cells, measurements, and traffic records are involved. Network operators need methods that can organize the data and highlight records that may require attention. Measurements such as RSRP and RSRQ describe signal strength and quality. SINR describes the signal compared with interference and noise. CQI, MCS, and BLER provide information about channel quality, transmission settings, and errors. When these values are studied together with traffic behaviour, they may support earlier and more informed network monitoring.

The dataset paper explains that the radio telemetry and higher layer traffic were collected at the same time in a live 5G O RAN testbed. It also reports that one experiment produced millions of higher layer packet records while the radio telemetry was sampled at a lower rate [2]. This combination is useful for a big data course because it includes both volume and variety. The project can study how to reduce, aggregate, and connect the large traffic data with the smaller but meaningful 5G measurements.

3. Dataset Description

The selected dataset is the NetsLab 5G Open RAN Intrusion Detection Dataset, referred to in this proposal as NetsLab 5G O RAN IDD. NetsLab describes it as a real world dataset collected with a physical 5G Open RAN testbed. It includes higher layer network traffic and lower layer radio and device measurements [1]. The dataset was created using normal user activity and several controlled abnormal traffic scenarios. This provides labels that can support basic prediction experiments without requiring the project to create a new 5G testbed.

Data group	Example fields	Possible use in the project
Radio measurements	RSRP, RSRQ, RSSI, SINR, CQI, MCS, BLER, power information, UE ID, cell ID	Study radio quality and add 5G context to each network state
Network flows	Protocol, duration, packet counts, byte counts, service, connection state	Create traffic summaries and input features for analysis
Packet captures	Raw packet data from selected scenarios	Optional deeper processing when time and computing cost allow
Labels and scenarios	Normal traffic and different abnormal traffic categories	Support binary or multiclass prediction experiments

Table 1. Main data groups in the selected dataset

4. General Project Aim and Questions

The general aim is to design and test a cloud based pipeline for large and mixed 5G network data. The work may examine how radio measurements and network traffic can be prepared for analysis and used in a simple prediction task. The following questions guide the proposal without limiting the final implementation.

1. **Data preparation.** How can large network flow data and 5G radio measurements be cleaned, grouped, and stored in a form that is easier to analyze?
2. **Data connection.** How can records from different layers be connected by time, user, cell, or scenario when the sampling rates are different?
3. **Prediction.** Can a simple model use radio and traffic features to estimate whether the current network state is normal or abnormal?
4. **Cloud value.** What practical benefits and costs appear when the same processing task is moved from a local computer to AWS?

5. Proposed Pipeline and Architecture

The architecture is designed as a flexible example rather than a fixed production system. Amazon S3 can act as the main cloud storage area. AWS describes S3 as object storage that can support large data and analytics workloads [3]. Amazon EMR may be used for Apache Spark and Hadoop processing. AWS describes EMR as a managed platform for running big data frameworks such as Spark and Hadoop [4]. A smaller setup using one EC2 instance or a local Spark environment may be used when cost or setup time is a concern.

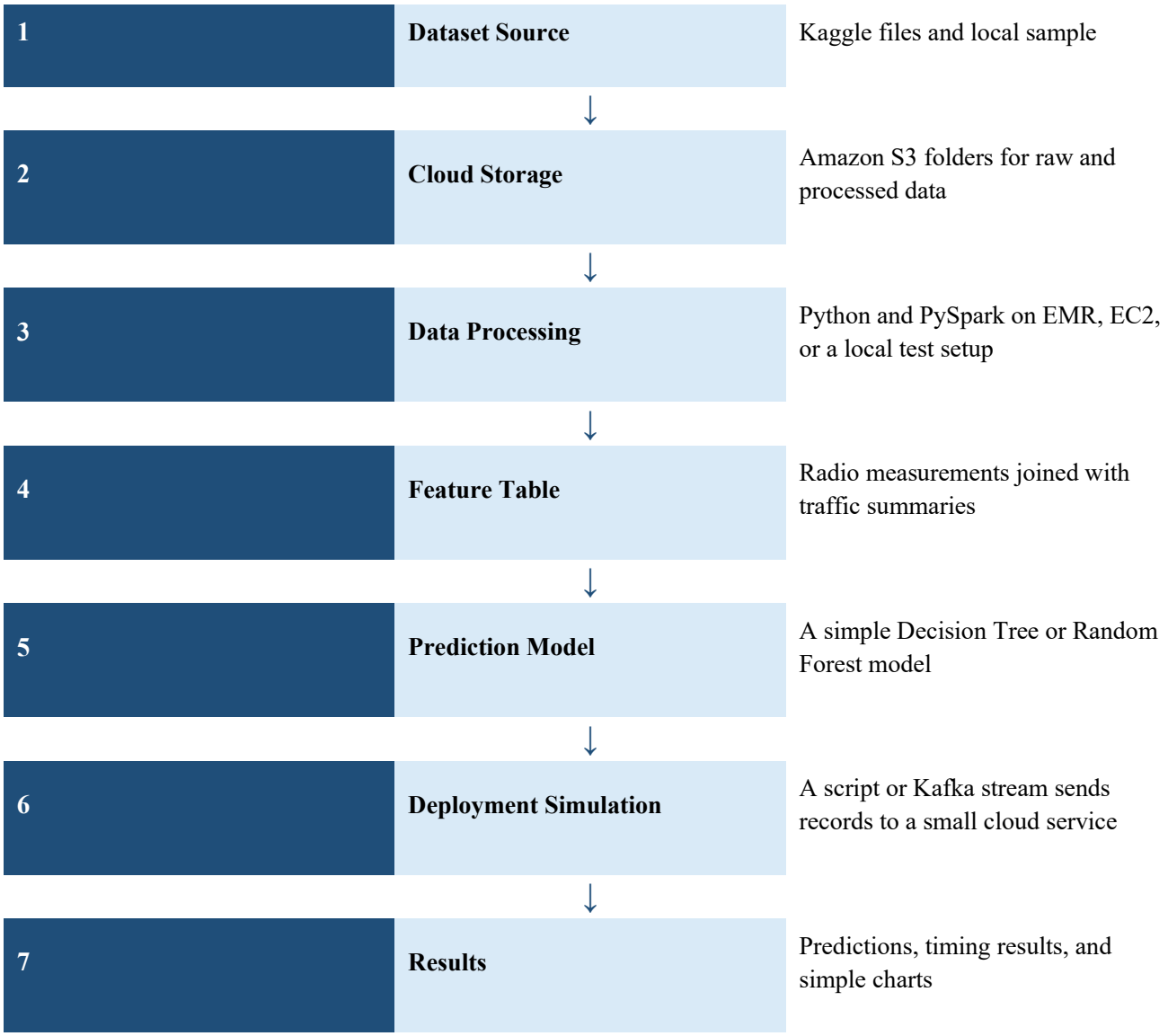


Figure 1. Flexible AWS architecture for the proposed project

6. Data Processing Plan

The first processing step may use the structured radio and flow files rather than the full raw packet captures. Python can be used to inspect columns, data types, missing values, and labels. PySpark can then read larger files and perform the same work across partitions. Apache Spark is designed for large scale data analytics and supports Python through PySpark [5]. Hadoop may provide the distributed storage and cluster concepts used by the course, either directly through HDFS or through services available in Amazon EMR [4] [6].

Stage	Example activity
Read and inspect	Load selected files and confirm columns, labels, and time formats
Clean	Handle missing values, invalid values, repeated records, and inconsistent data types
Aggregate	Create traffic summaries for a selected time window and user or scenario
Join	Connect radio values with traffic summaries using available time and identity fields
Transform	Store the prepared table as compressed Parquet files in Amazon S3
Analyze	Calculate summary statistics and prepare a simple prediction model

7. Possible Prediction Task

The simplest useful prediction is a binary network state. The output can be normal or abnormal. This target is easy to explain and matches the labels available in the dataset. The input may include selected radio measurements, traffic summaries, or both. A Decision Tree is easy to interpret, while a Random Forest can provide a stronger baseline. The exact model can be selected after checking the final feature table.

The prediction can act as a simple early warning. Instead of asking a person to review every network record, the system can identify records that look different from normal behavior. This may help a network team focus on the times, users, or cells that need more attention. It can also show whether radio measurements add useful information beyond packet and flow statistics.

Model version	Example input	Purpose
Radio only	RSRP, RSRQ, RSSI, SINR, CQI, MCS, BLER	Check what can be learned from 5G measurements
Traffic only	Packet count, byte rate, duration, protocol, connection count	Create a baseline using network behaviour
Combined	Radio features and traffic features	Check whether using both layers gives a clearer prediction

8. Proposed Tools and Programming Language

Cloud infrastructure is useful because data storage, processing, and model access can be placed in one environment. Amazon S3 can store raw and prepared files. Amazon EMR can provide temporary Spark and Hadoop resources when a distributed test is needed. The resources can be started for a short experiment and stopped afterward. This is different from buying or maintaining a permanent local cluster.

Python is the main programming language because it can be used for data preparation, PySpark jobs, model training, AWS access, and a small prediction service. Python also has a large standard library and a wide range of data and machine learning packages [8]. The exact set of tools may change after the first week. The following table shows possible choices and keeps advanced tools optional.

Tool	Possible role	Status in the proposal
Python	Main language for scripts, data checks, models, and cloud access	Core
PySpark and Apache Spark	Distributed cleaning, aggregation, joins, and analysis	Core or small local test
Amazon S3	Cloud storage for raw and processed data	Core
Amazon EMR	Managed Spark and Hadoop environment	Possible cloud processing choice
Hadoop and HDFS	Distributed storage and processing concepts	Possible through EMR or local setup
Apache Kafka	Replay records as a stream and simulate incoming measurements	Possible data processing tool
scikit learn or Spark MLlib	Decision Tree or Random Forest model	One library can be selected
EC2, Lambda, or a small API service	Host a simple prediction endpoint	Optional deployment choice
Jupyter Notebook	Explore data and document experiments	Helpful

9. Four Week Milestone Plan

Time	Main focus	Flexible activities
Week 1	Dataset understanding and scope	Download or inspect a local sample, identify radio and traffic files, review labels, and choose a manageable project subset. Create the AWS account structure and basic S3 folders if suitable.
Week 2	Data preparation and cloud processing	Develop Python and PySpark steps for cleaning, aggregation, and joining. Test locally first, then try a selected job with AWS resources. Save processed data in Parquet format.
Week 3	Prediction and deployment simulation	Train one simple model and compare a small number of feature choices. Save the model. Test a basic cloud endpoint or a local service that represents the deployment.
Week 4	Evaluation and presentation	Complete the final report and presentation.

References

[1] UCD NetsLab. NetsLab 5G Open RAN Intrusion Detection Dataset. Dataset information and Kaggle access. Accessed July 2026. NetsLab dataset page

[2] A. Civciss, V. Ravihansa, F. Abed Zadeh, C. Sandeepa, and M. Liyanage. Silent Signals, Loud Threats: Using dApps for Radio Signal Intelligence based Intrusion Detection in 5G O RAN. IEEE GLOBECOM, 2025.

[3] Amazon Web Services. What is Amazon S3. AWS documentation. Accessed July 2026. AWS S3 documentation

[4] Amazon Web Services. What is Amazon EMR. AWS documentation. Accessed July 2026. AWS EMR documentation

[5] Apache Software Foundation. Apache Spark documentation. Accessed July 2026. Apache Spark

[6] Apache Software Foundation. Apache Hadoop documentation. Accessed July 2026. Apache Hadoop

[7] Apache Software Foundation. Apache Kafka documentation. Accessed July 2026. Apache Kafka

[8] Python Software Foundation. Python documentation. Accessed July 2026. Python documentation

End of Proposal